water & sanitation
Department:
Water and Sanitation
REPUBLIC OF SOUTH AFRICA

NATIONAL INTEGRATED WATER INFORMATION SYSTEM

Custom NIWIS Report

Generated	02 Nov 2026, 12:09 (SAST)
Data as of	02 Nov 2026
Area	South Africa (national)
Period	12M
Issued by	Department of Water and Sanitation, Republic of South Africa

Rainfall Status Monitoring

Live · as of Aug 2026

Total Rainfall
07 Sep 2025 – 06 Sep 2026

702 mm ↗ +113 mm

vs same period last year



Rainfall Anomaly
vs 2000–2026 normal (533 mm)

32 %

Heavy Rain Days (R20mm)
days ≥ 20 mm, area-weighted district mean

6 days

Max Daily Rainfall
Vhembe, 12 Jan 2026

158 mm

Consecutive Dry Days
current spell of days < 1 mm to 06 Sep

1 days

SPI-6
Aug 2026 (latest complete month)

0.44 No drought

SPEI-6
Aug 2026 (latest complete month)

0.36 No drought

Active Rainfall Stations

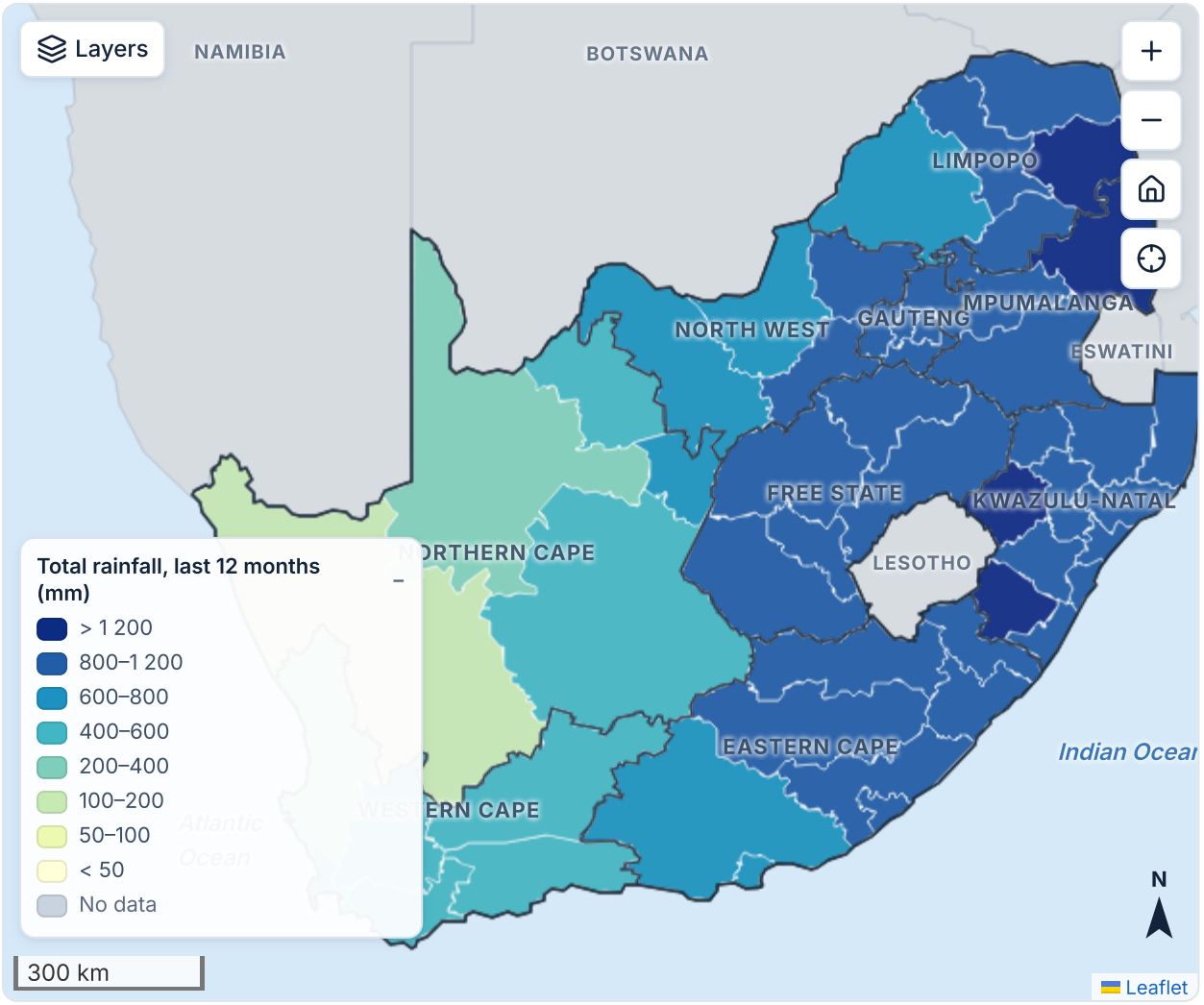
—

No rainfall-station data in the database (data are per district municipality)

Rainfall Distribution Map – South Africa

● Live · as of Sep 2026

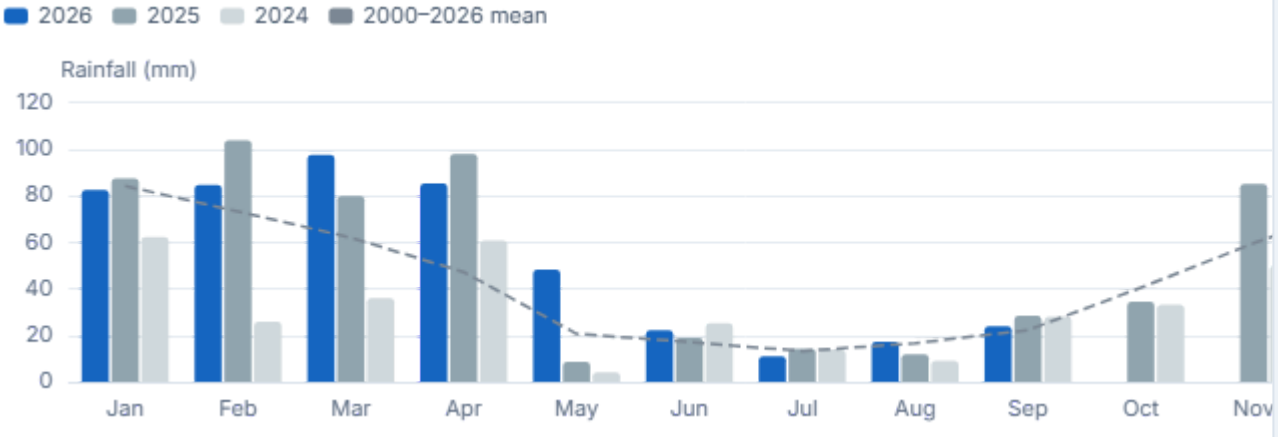
Total rainfall | Rainfall anomaly | Monthly rainfall | Mean annual (MAP) | SPI-1 | SPI-3 | SPI-6 | SF



Monthly Rainfall – Current vs Previous Years

Normal: 2000–2026 mean (data start in 2000). Sep 2026 is to 06 Sep.

● Live · as of Sep 2026

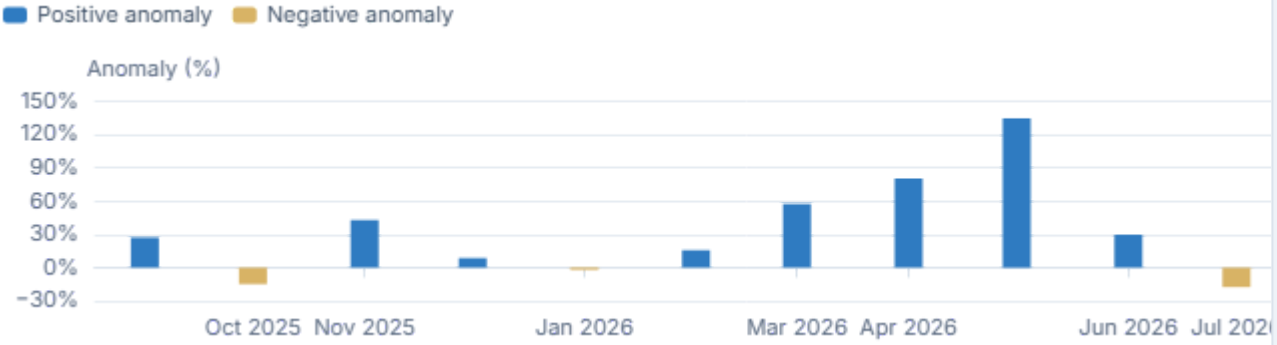


Rainfall Anomaly (vs 2000–2026 mean)

Complete months only; anomaly vs the 2000–2026 monthly mean.

● Live · as of Aug 2026

1-Month | 3-Month | 6-Month | 12-Month



Rainfall by Province (12 Months)

● Live · as of Sep 2026

#	Province	Total (12 m) (mm)	Anomaly	Status
1	Mpumalanga	1 175	↗ +42%	No drought
2	Gauteng	1 053	↗ +43%	No drought
3	KwaZulu-Natal	1 033	↗ +8%	No drought
4	Limpopo	950	↗ +74%	No drought
5	Free State	945	↗ +39%	No drought
6	Eastern Cape	822	↗ +15%	No drought
7	North West	746	↗ +37%	No drought
8	Western Cape	393	↗ +8%	No drought
9	Northern Cape	375	↗ +39%	No drought

Top 5 Wettest & Driest Locations

● Live · as of Sep 2026

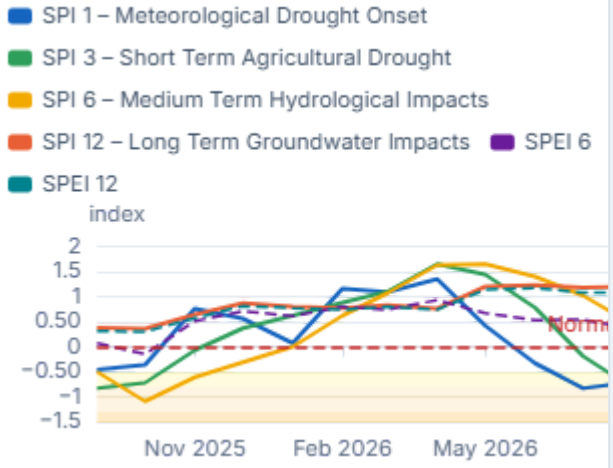
Wettest | Driest

#	Location	Province	Total (12 m) (mm)
1	Mopani	LP	1 255
2	Ehlanzeni	MP	1 244
3	uThukela	KZN	1 233
4	Harry Gwala	KZN	1 210
5	Thabo Mofutsanyana	FS	1 172
6	uMgungundlovu	KZN	1 165
7	Nkangala	MP	1 141
8	City of Ekurhuleni	GP	1 132
9	Gert Sibande	MP	1 131
10	Amajuba	KZN	1 125

SPI and SPEI Trends

Area-weighted mean of district indices (approximation of a regional SPI). Complete months only.

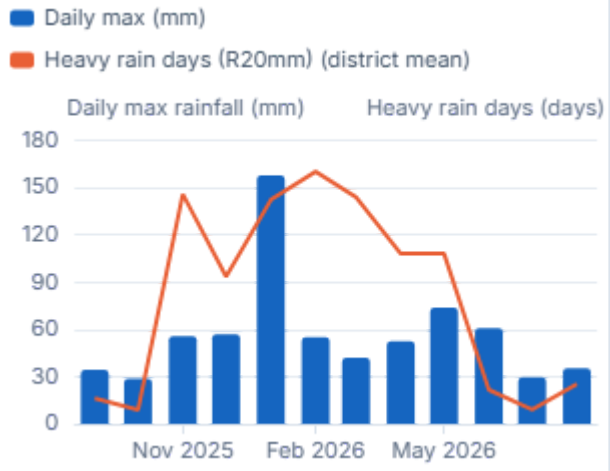
● Live · as of Aug 2026



Rainfall Intensity and Extreme Events

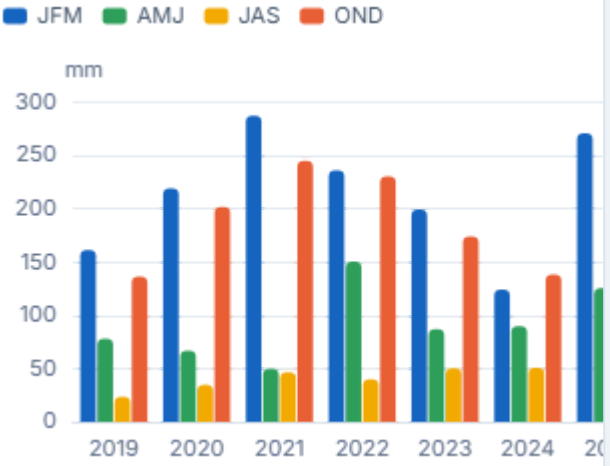
Daily max = highest district daily total in the scope; R20mm days = area-weighted district mean.

● Live · as of Aug 2026



Seasonal Rainfall Distribution

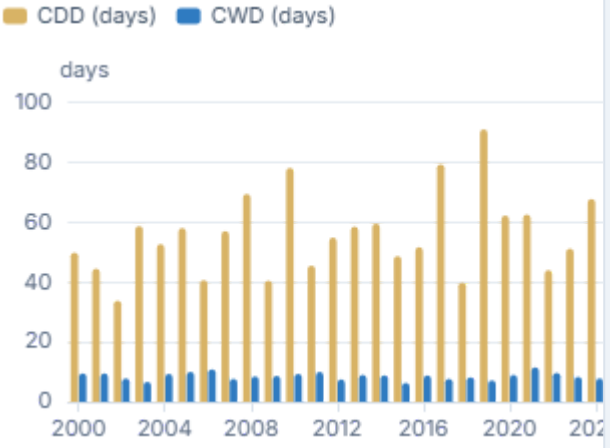
● Live · as of Dec 2025



Consecutive Dry and Wet Days

Annual ETCCDI indices, full years only (area-weighted district mean).

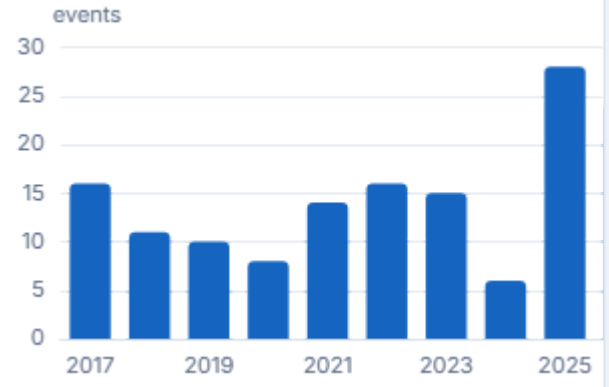
● Live · as of Dec 2025



Extreme Rainfall Events

● Live · as of Sep 2026

Event count | Maximum 1-day rainfall



Consecutive Dry Days (Current)

Consecutive dry days (current, to 06 Sep) (days)

● Live · as of Sep 2026

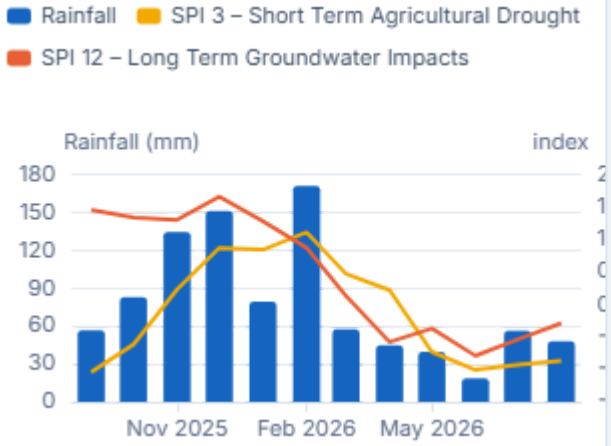


< 7 | 7–15 | 15–30 | 30–45 | 45–60 | > 60

Rainfall Indices Time Series (Selected District)

Alfred Nzo

● Live · as of Aug 2026



District values (NIWIS data are per district municipality; no stations).

Rainfall Data Sources and Coverage

● Live · as of Sep 2026

Source	Stations	Resolution	C
District daily rainfall	—	52 districts, daily	
CHIRPS daily rainfall estimate	—	52 districts, daily	
TAMSAT rainfall estimate (mm per dekad, one value per month)	—	52 districts, monthly	
SPI (1–24 months)	—	52 districts, monthly	
SPEI (1–24 months)	—	52 districts, monthly	
ETCCDI rainfall		52 districts	

Scroll sideways for more columns →

Not available: Stations *(No station data in the database)*

Rainfall Statistics (Selected Area)

● Live · as of Dec 2025

Variable	Value (2025)	Climatology (2000–2025)
Annual rainfall	653 mm	528 m
Rainy days (≥ 1 mm)	99 days	88 da
Heavy rain days (R20mm)	5.0 days	3.1 da
Max 1-day rainfall (Rx1day)	35.1 mm	30.4 m
Consecutive dry days (CDD)	42 days	55 da

Scroll sideways for more columns →

Alerts & Early Warnings

● Live

View all →

⚠ D1 Moderate Drought Alfred Nzo	01 Aug	Moderate
⚠ D1 Moderate Drought Amajuba	01 Aug	Moderate
⚠ D1 Moderate Drought Harry Gwala	01 Aug	Moderate
⚠ D1 Moderate Drought King Cetshwayo	01 Aug	Moderate
⚠ D1 Moderate Drought O.R. Tambo	01 Aug	Moderate

Rainfall Outlook – Month 1

Rainfall outlook +1 month (anomaly, % of the same-month mean) (%) – Sep 2026

● Live · as of Aug 2026



< −50 · −50 to −25 · −25 to −10 · −10 to +10 · +10 to +25 · +25 to +50 · > +50

Rainfall Outlook – Month 2

Rainfall outlook +2 months (anomaly, % of the same-month mean) (%) – Oct 2026

● Live · as of Aug 2026



< −50 · −50 to −25 · −25 to −10 · −10 to +10 · +10 to +25 · +25 to +50 · > +50

Rainfall Outlook – Month 3

Rainfall outlook +3 months (anomaly, % of the same-month mean) (%) – Nov 2026

● Live · as of Aug 2026



< −50 · −50 to −25 · −25 to −10 · −10 to +10 · +10 to +25 · +25 to +50 · > +50

Database variables

Every table and column of the national database for this page, with the database's own names and units.

Spec coverage

● Live · as of Sep 2026

Rainfall: 4 tables · 41 columns · 34 live · 5 withheld · 2 duplicates Show every column →

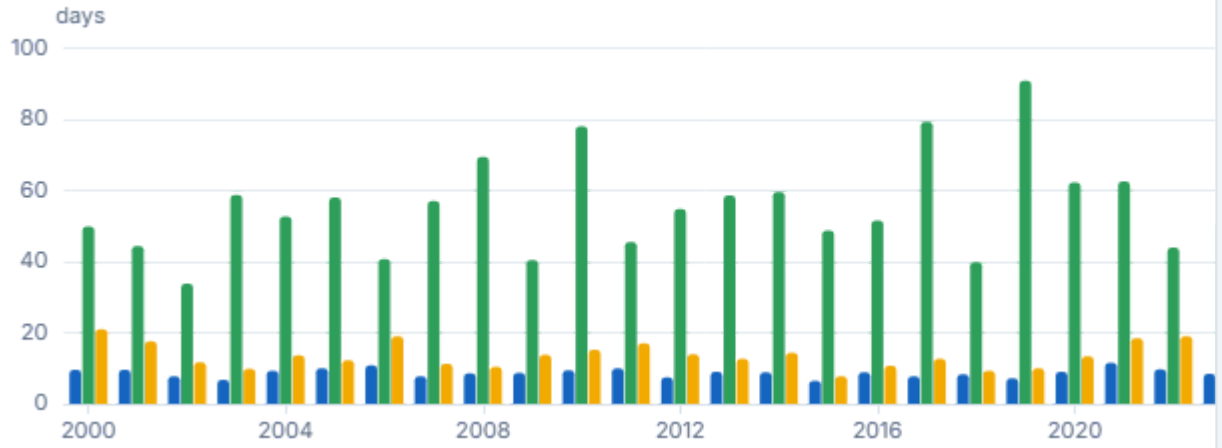
Live columns are plotted below. Withheld columns failed a check or hold placeholder values (reason given); duplicates repeat another column.

Wet Dry Spells §4.2 `4_2_wet_dry_spells_master` 8 live · 0 withheld · 0 duplicates data to 2025

Wet Dry Spells · CWD / CDD / R10mm / R20mm (days) · §4.2

● Live · as of Dec 2025

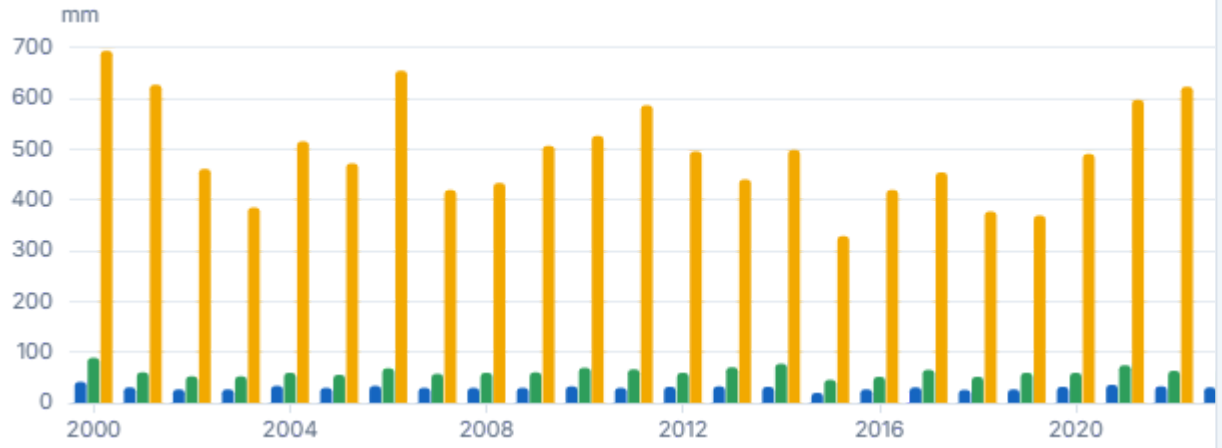
CWD (days) | CDD (days) | R10mm (days) | R20mm (days) | All | Reset



Wet Dry Spells · Rx1day / Rx5day / PRCPTOT (mm) · §4.2

● Live · as of Dec 2025

Rx1day (mm) | Rx5day (mm) | PRCPTOT (mm) | All | Reset



Wet Dry Spells · SDII (mm/wet day) · §4.2

Live · as of Dec 2025

SDII (mm/wet day)



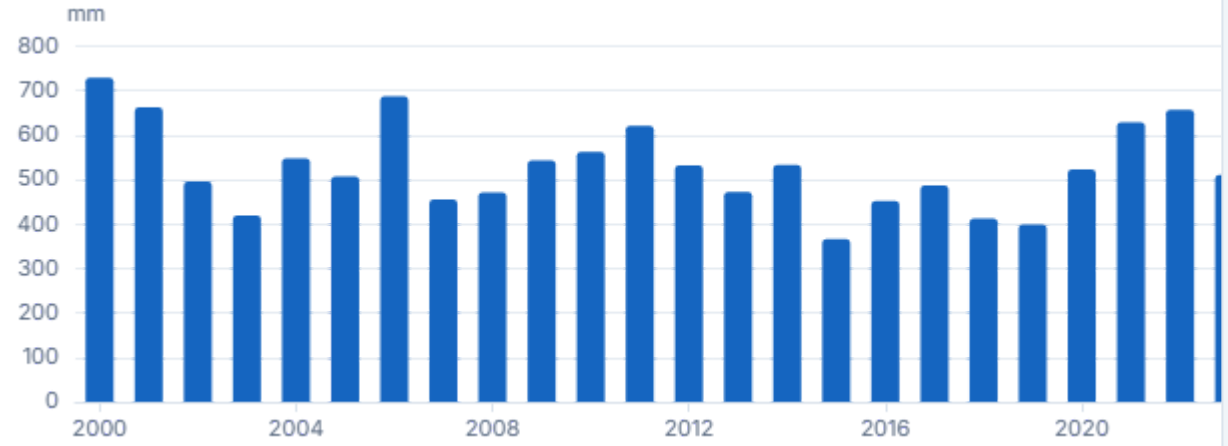
Mean Annual Precipitation §4.3 `4_3_mean_annual_precipitation_master`

6 live · 5 withheld · 0 duplicates data to 2025

Mean Annual Precipitation · MAP (mm) · §4.3

Live · as of Dec 2025

MAP (mm)

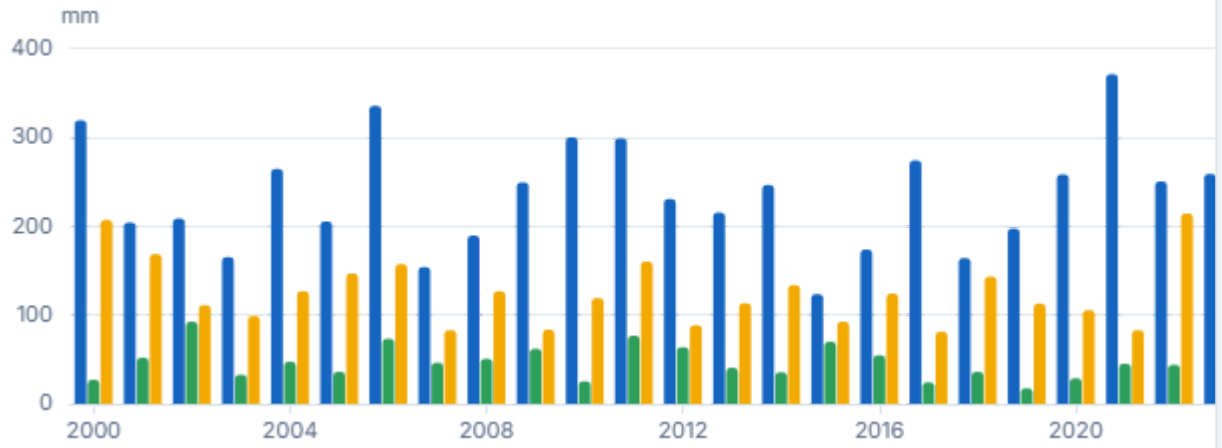


Mean Annual Precipitation · Precipitation (mm) · §4.3

Live · as of Dec 2025

Precip DJF (mm) | Precip JJA (mm) | Precip MAM (mm) | Precip SON (mm)

All Reset



Mean Annual Precipitation · Nino34 Annual Mean · §4.3

Live · as of Dec 2025

Nino34 Annual Mean



Mean Annual Precipitation · Precip CV (%) · §4.3

~~Precip CV (%)~~

Not yet available

Withheld: Precip CV (%) (computed over 2000–2026 including the partial year 2026 (Alfred Nzo 17.05 %; 14.70 % over 2000–2025))

Mean Annual Precipitation · MK Sens Slope (mm/year) · §4.3

~~MK Sens Slope (mm/year)~~

Not yet available

Withheld: MK Sens Slope (mm/year) (computed over 2000–2026 including the partial year 2026: Alfred Nzo −1.03 mm/yr stored, +0.22 mm/yr over the full years 2000–2025 (app.core.stats))

Mean Annual Precipitation · MK P Value · §4.3

~~MK P Value~~

Not yet available

Withheld: MK P Value (computed over 2000–2026 including the partial year 2026 (same test as MK_Sens_Slope_mm_yr))

Mean Annual Precipitation · MK Trend (class) · §4.3

~~MK Trend~~

Not yet available

Withheld: MK Trend (computed over 2000–2026 including the partial year 2026 (same test as MK_Sens_Slope_mm_yr))

Mean Annual Precipitation · ENSO Nino34 Correlation · §4.3

~~ENSO Nino34 Correlation~~

Not yet available

Withheld: ENSO Nino34 Correlation (includes 2026, whose Niño 3.4 value is −41.28: Alfred Nzo +0.48 stored, −0.43 over 2000–2025)

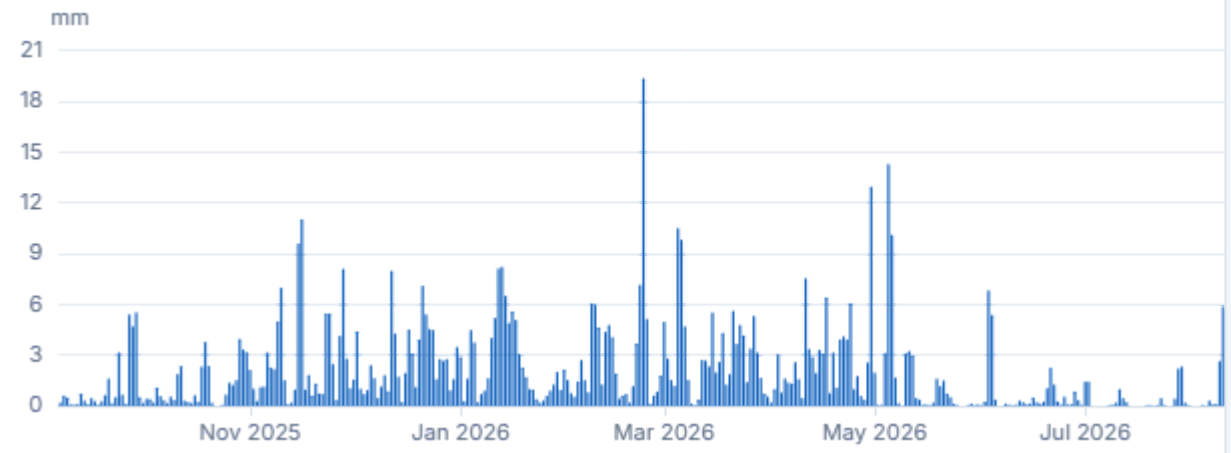
Core Rainfall Variables §6.1 `6.1_core_rainfall_variables_master`

12 live · 0 withheld · 2 duplicates data to Sep 2026

Core Rainfall Variables · Daily Rainfall Total (mm) · §6.1

● Live · as of Sep 2026

Daily Rainfall Total (mm)

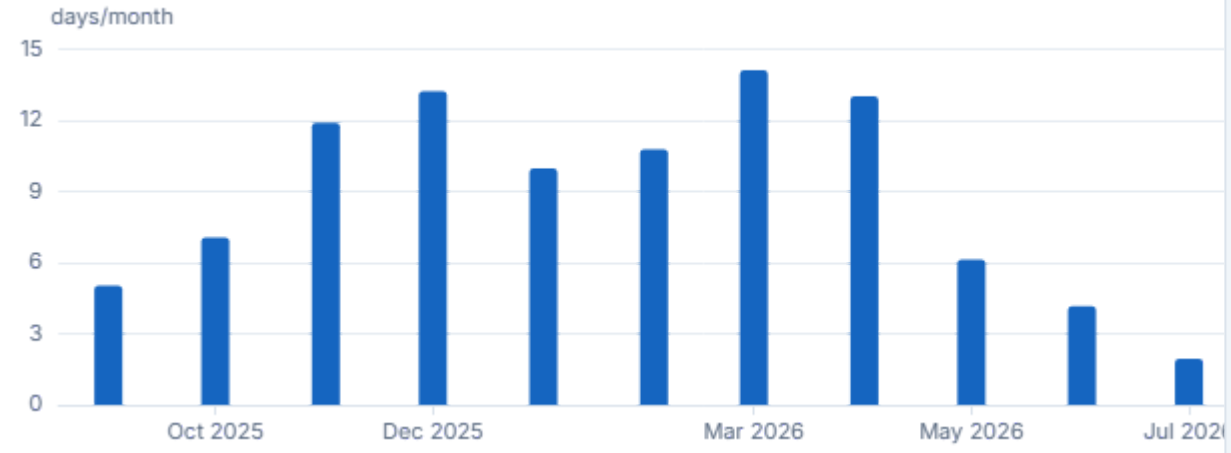


Also in the database: Gridded Rainfall Product 1km (mm) (identical to 6.1_core_rainfall_variables_master.Daily_Rainfall_Total_mm)

Core Rainfall Variables · Number of Rainy Days (days/month) · §6.1

● Live · as of Aug 2026

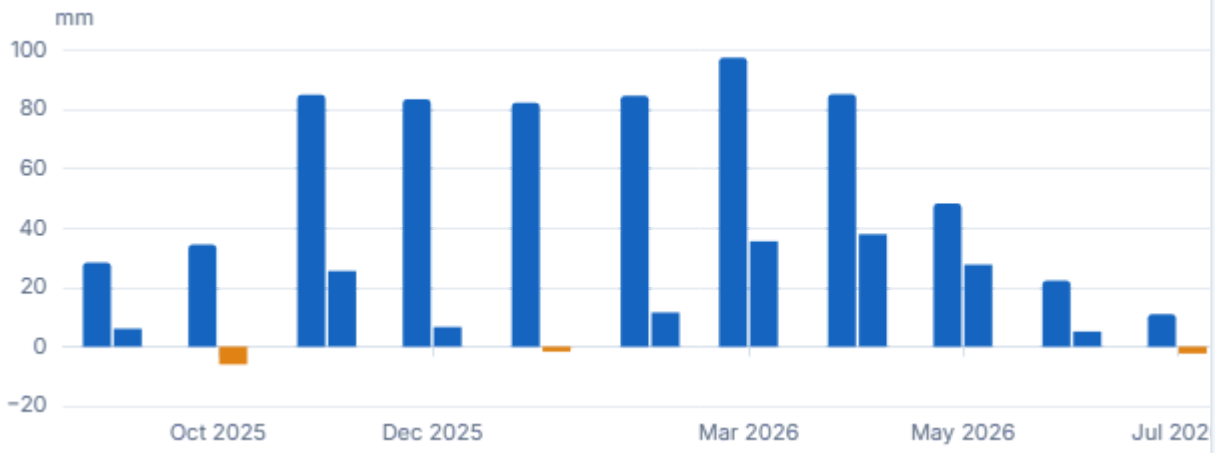
Number of Rainy Days (days/month)



Core Rainfall Variables · Monthly (mm) · §6.1

● Live · as of Aug 2026

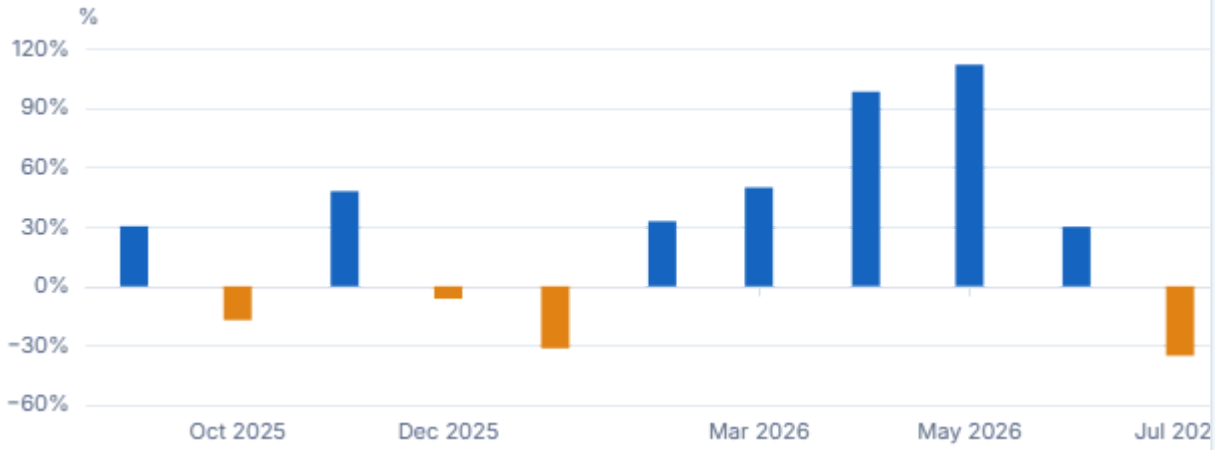
Monthly Total Precipitation (mm) | Monthly Anomaly (mm) | All | Reset



Core Rainfall Variables · Monthly Anomaly (%) · §6.1

● Live · as of Aug 2026

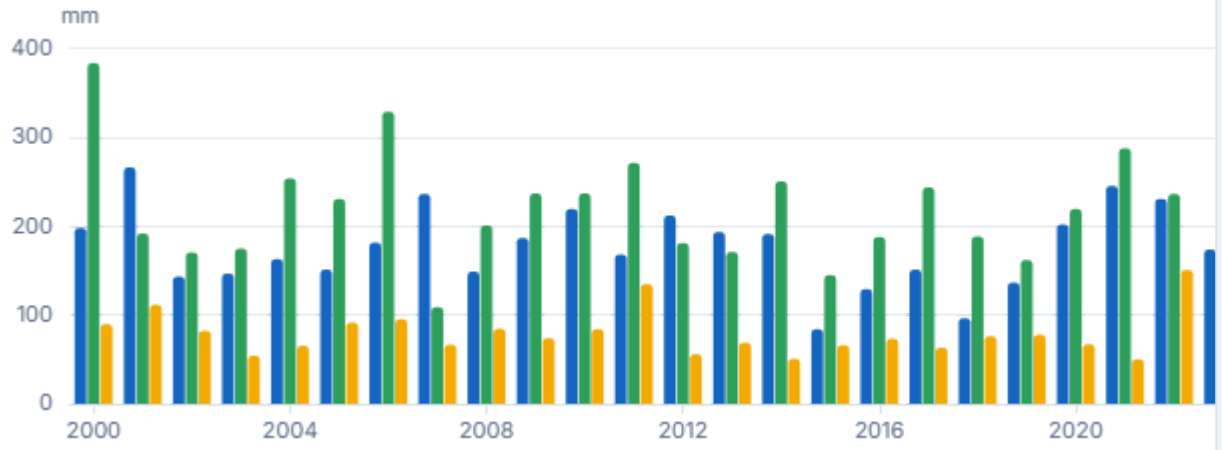
Monthly Anomaly (%)



Core Rainfall Variables · Seasonal Total (mm) · §6.1

● Live · as of Dec 2025

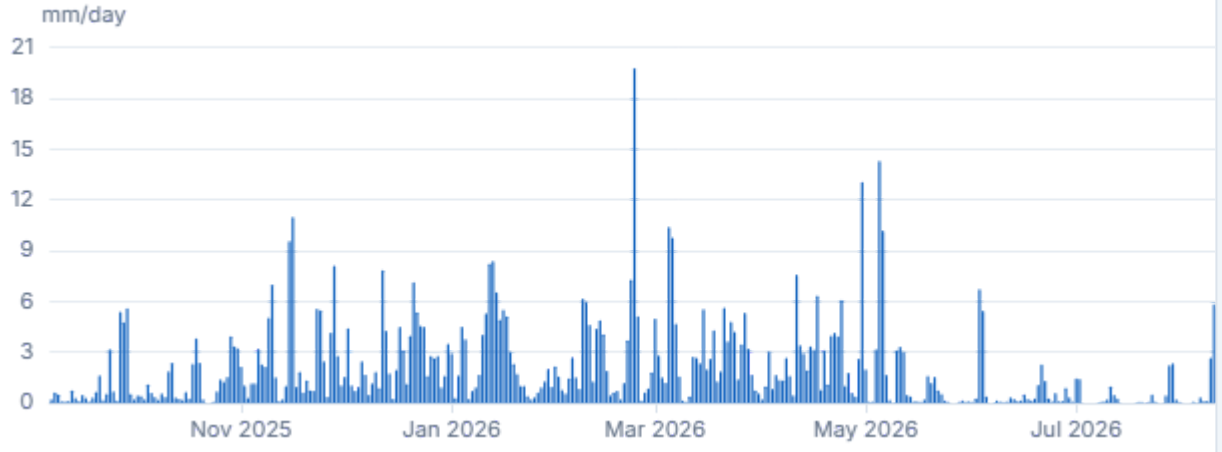
Seasonal Total OND (mm) | Seasonal Total JFM (mm) | Seasonal Total AMJ (mm) | Seasonal Total JAS (mm) | All | Reset



Core Rainfall Variables · CHIRPS Daily Rainfall Estimate (mm/day) · §6.1

● Live · as of Sep 2026

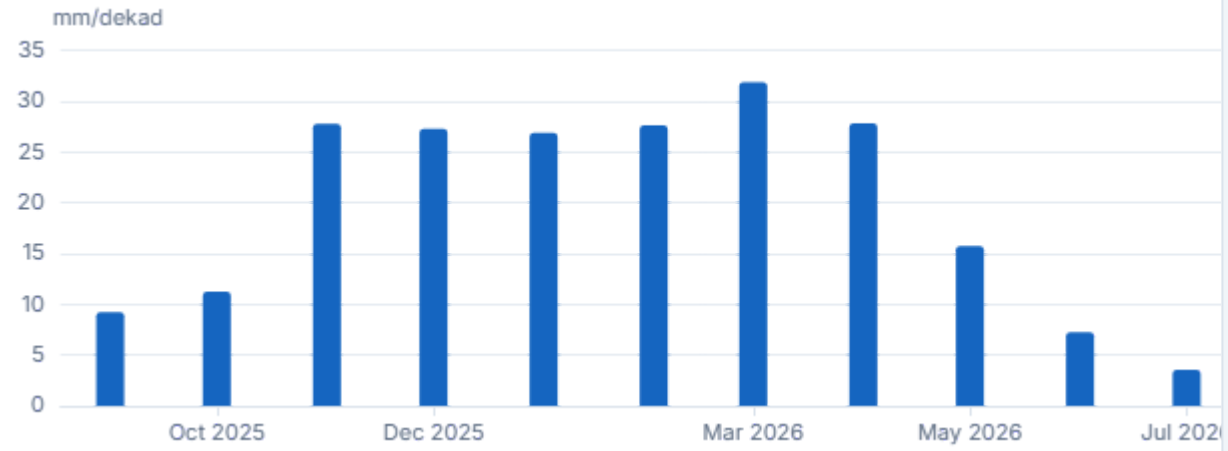
CHIRPS Daily Rainfall Estimate (mm/day)



Core Rainfall Variables · TAMSAT Satellite Rainfall Estimate (mm/dekad) · §6.1

Live · as of Aug 2026

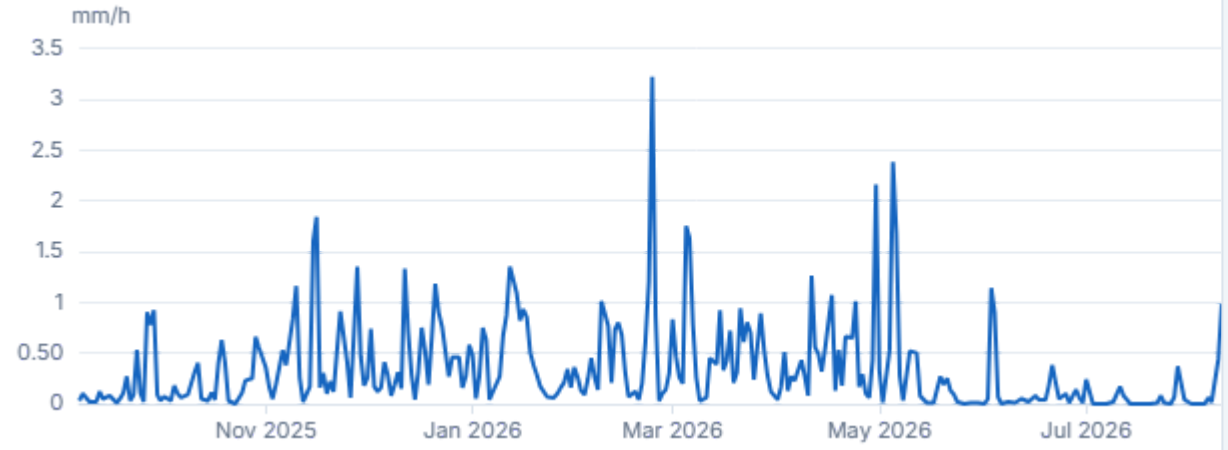
TAMSAT Satellite Rainfall Estimate (mm/dekad)



Core Rainfall Variables · Hourly Rainfall Intensity (mm/h) · §6.1

Live · as of Sep 2026

Hourly Rainfall Intensity (mm/h)



Also in the database: Annual Total Precipitation (mm) (identical to 4_3_mean_annual_precipitation_master.MAP_mm)

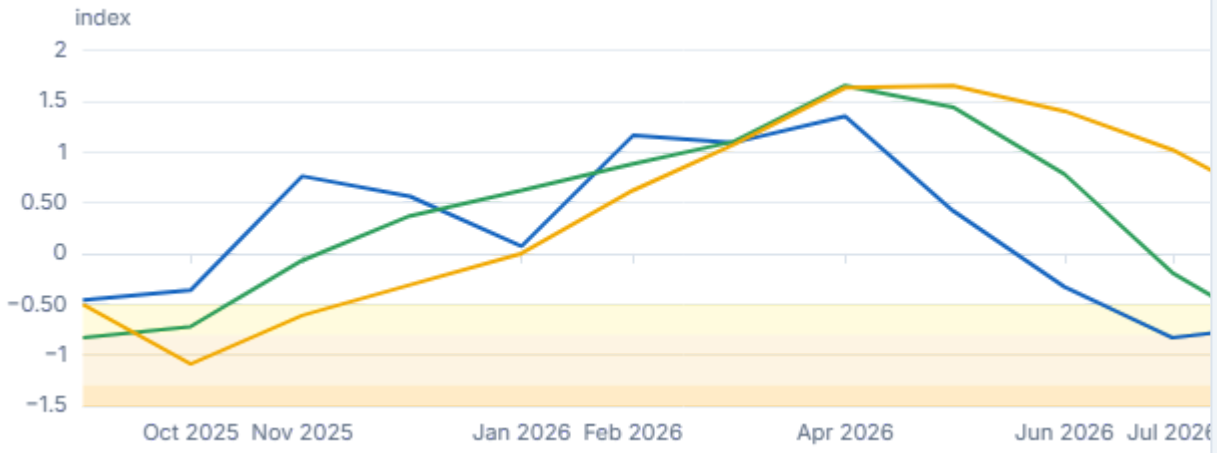
Rainfall Derived Drought Indices §6.2 `6.2_rainfall_derived_drought_indices_master`

8 live · 0 withheld · 0 duplicates data to Aug 2026

Rainfall Derived Drought Indices · SPI / SPEI / RAI (index) · §6.2

● Live · as of Aug 2026

SPI 1 – Meteorological Drought Onset | SPI 3 – Short Term Agricultural Drought | SPI 6 – Medium Term Hydrological Impacts | SPI 12 – Long Term Groundwater Impacts | SPI 24 – Very Long Term Drought | SPEI 3 – Agricultural Drought ET | SPEI 12 – Annual Water Balance Drought | RAI – Rainfall Anomaly Index | All | Reset



Water Quality Monitoring

● Live · as of Oct 2026

Monitoring Stations

—

No coordinates in the database: every §8–§12 table is keyed by district municipality (no stations, boreholes, dams or sampling points with locations), so point layers cannot be drawn

Compliance Rate

Oct 2026: share of 14 parameters × 52 districts within guideline

94 %

Districts Exceeding Limits

at least one parameter outside its guideline

33 districts

Average WQI

CCME WQI, 12 months (area-weighted)

91 Excellent

Districts in Poor Quality

WQI below 50 (Poor or Very Poor)

0 districts

Dams with Good Quality

—

No water-body type (river, dam, wetland, groundwater) in the database: one record per district

Groundwater Quality

—

No water-body type (river, dam, wetland, groundwater) in the database: one record per district

High-Risk Districts

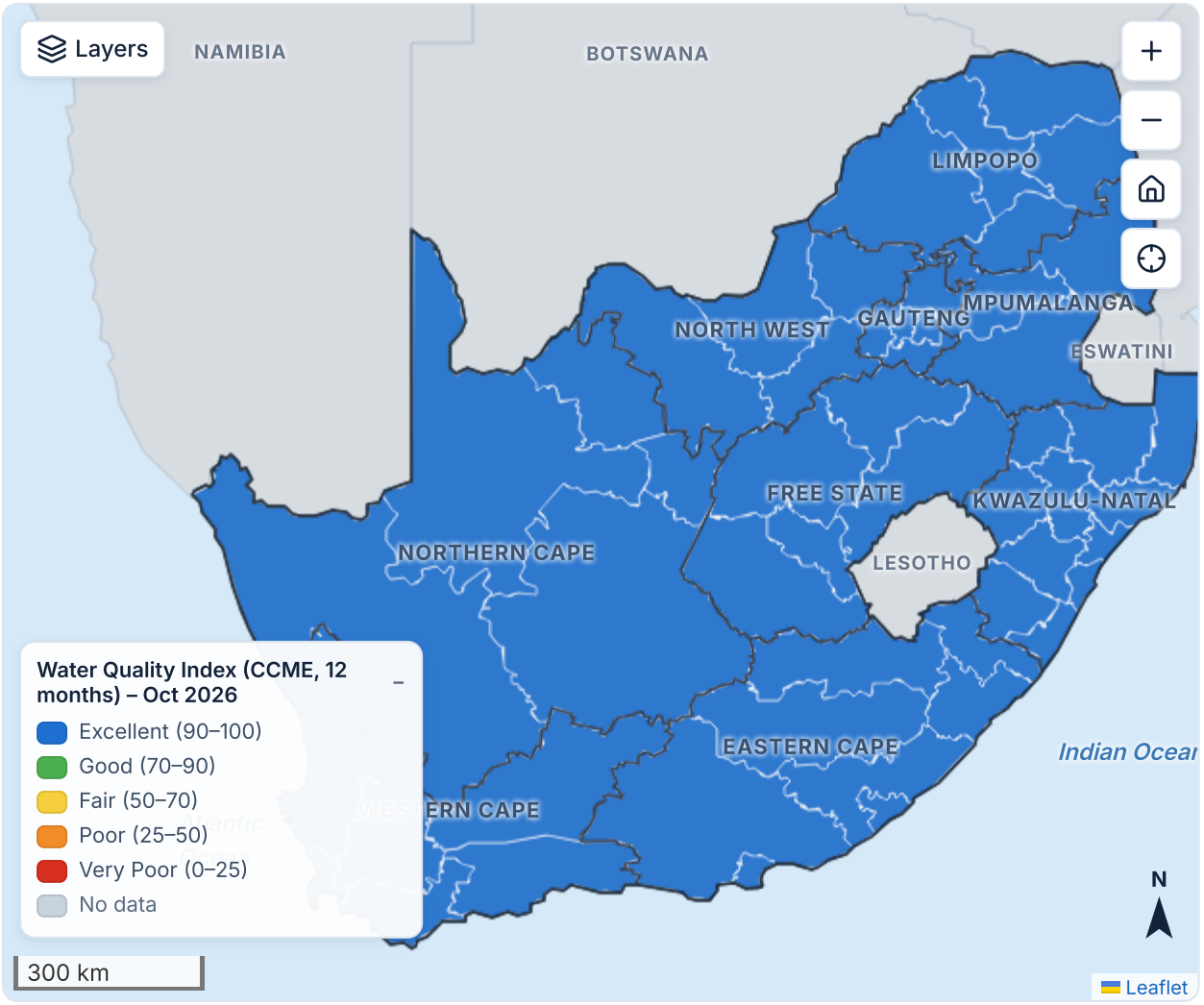
WQI below 25 (Very Poor)

0 districts

Water Quality Index (WQI) by District

● Live · as of Oct 2026

Water Quality Index (WQI) | pH | Dissolved oxygen | Electrical conductivity | Turbidity | Nitrate | P

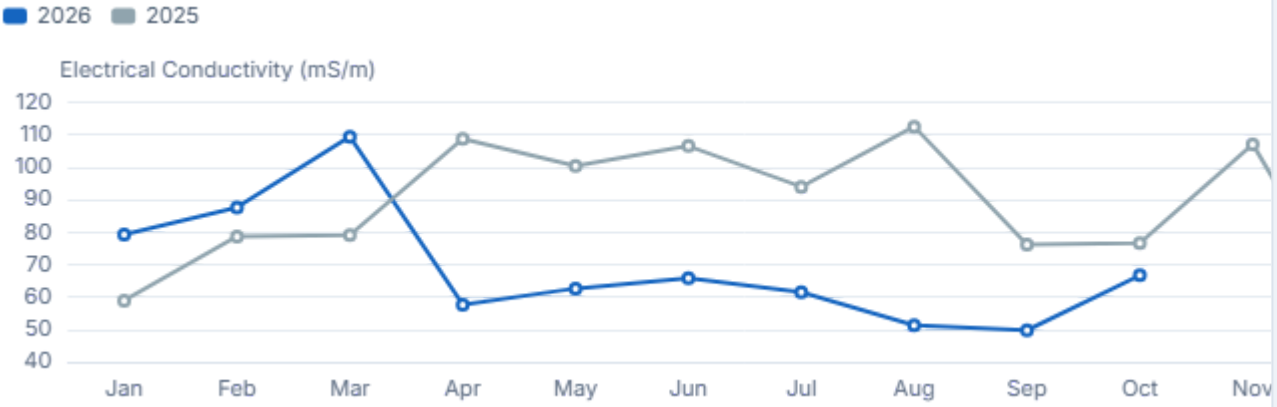


Key Water Quality Trends (Selected District)

[District: Alfred Nzo]

● Live · as of Oct 2026

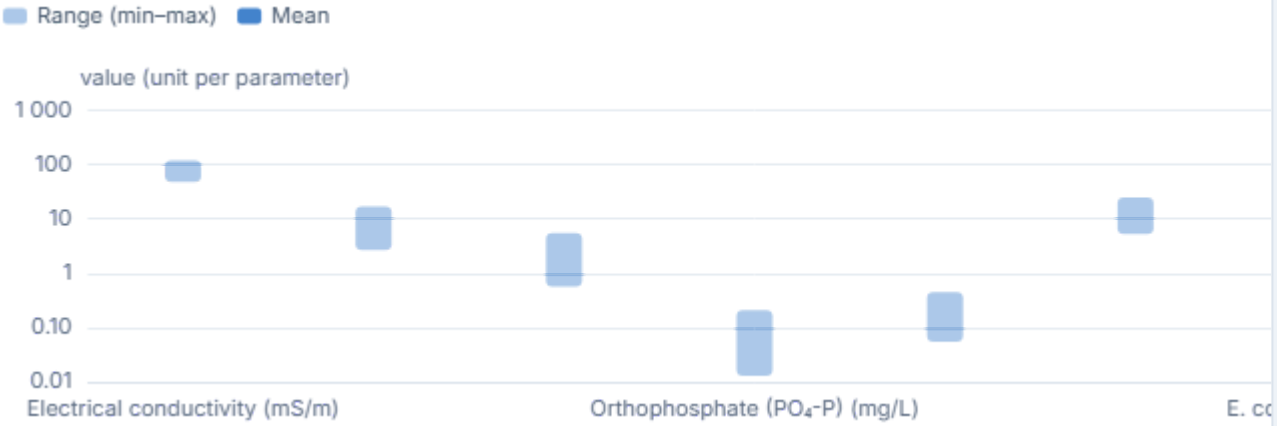
EC | pH | Turbidity | DO | BOD | COD | Nitrate | Phosphate | E. coli



Monthly values as stored (district value); Guidelines: resource quality objectives for domestic use (DWAF 1996 TWQR / SANS 241), thresholds.json `wq_guidelines`.

Water Quality by Parameter (Range, Log Scale)

● Live · as of Oct 2026



Top / Poorest Districts by WQI

● Live · as of Oct 2026

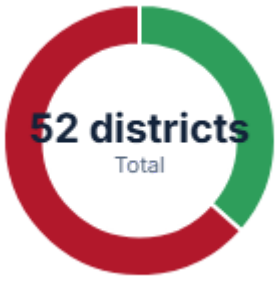
Poorest | Best

#	District	River	WQI	Status	vs 12 months ago
1	West Coast	—	90	Excellent	↘ −1.0
2	O.R. Tambo	—	91	Excellent	↘ −1.0
3	Xhariep	—	91	Excellent	↘ −0.1
4	Garden Route	—	91	Excellent	↘ −1.0
5	Lejweleputswa	—	91	Excellent	↘ −0.5
6	Vhembe	—	91	Excellent	↘ −0.5
7	ZF Mgcawu	—	91	Excellent	↘ −0.1
8	Waterberg	—	91	Excellent	↘ −0.8
9	Sedibeng	—	91	Excellent	↘ −0.4
10	uMgungundlovu	—	91	Excellent	↘ −0.5

Not available: River *(No water-body type (river, dam, wetland, groundwater) in the database: one record per district)*

Water Quality Compliance

● Live · as of Oct 2026

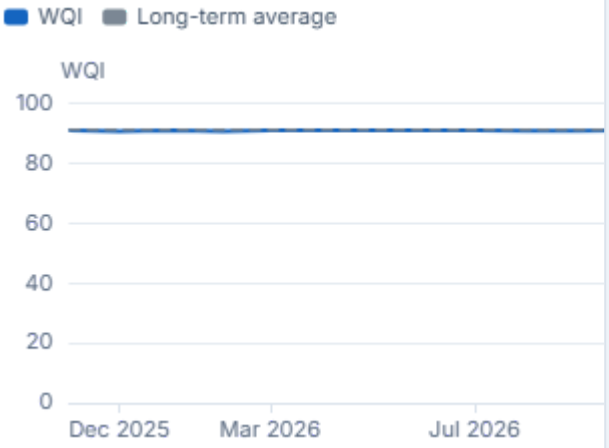


Compliant	19 districts	37%
Not compliant	33 districts	63%

WQI Time Series (Selected District)

[District: Alfred Nzo]

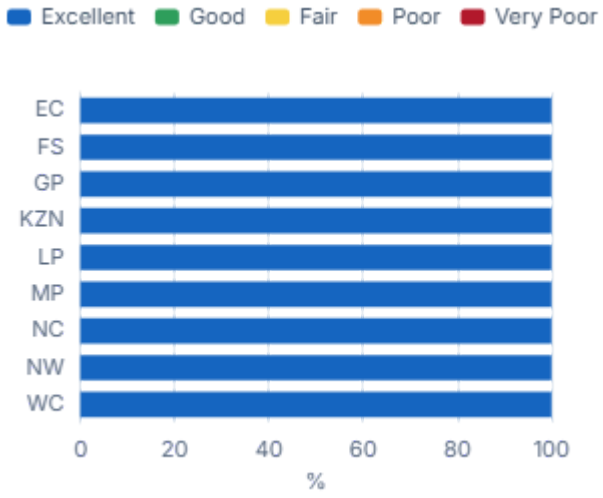
● Live · as of Oct 2026



CCME Water Quality Index (2001) of the trailing 12 months of 14 parameters against their guidelines; Guidelines: resource quality objectives for domestic use (DWAF 1996 TWQR / SANS 241), thresholds.json `wq_guidelines`.

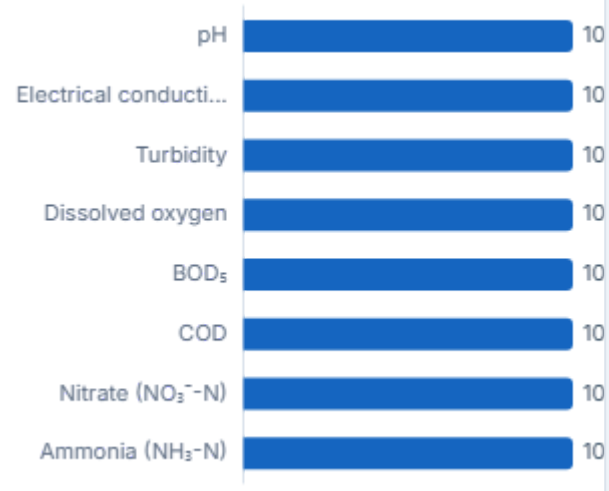
Water Quality Status by Province

● Live · as of Oct 2026



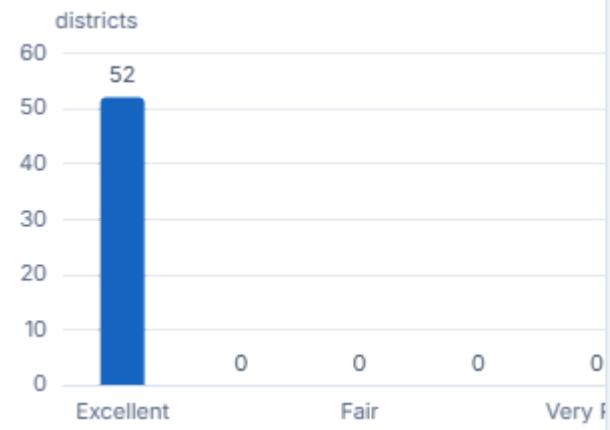
Compliance by Parameter

● Live · as of Oct 2026



Water Quality Index Distribution

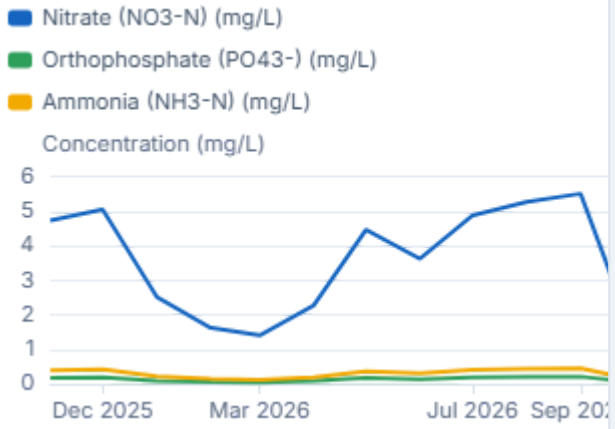
● Live · as of Oct 2026



Nutrients (Selected District)

[District: Alfred Nzo]

● Live · as of Oct 2026

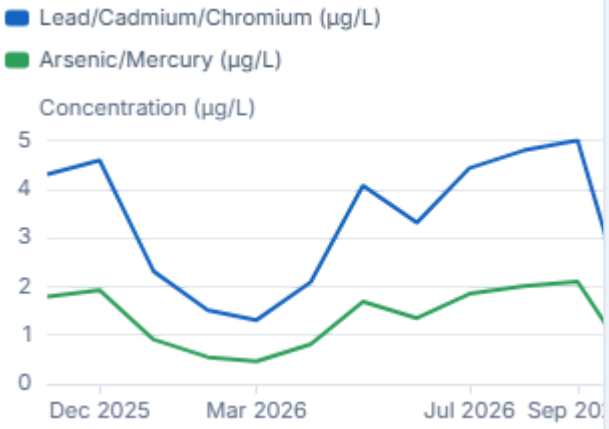


Monthly values as stored in 12.2 / 12.3

Metals (Selected District)

[District: Alfred Nzo]

● Live · as of Oct 2026

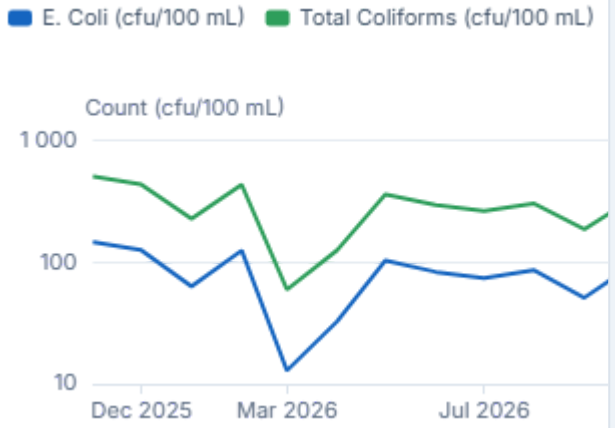


Monthly values as stored in 12.2 / 12.3 (Iron/Manganese is in mg/L: it is in the database view)

Microbiological – E. coli (Selected District)

[District: Alfred Nzo]

● Live · as of Oct 2026

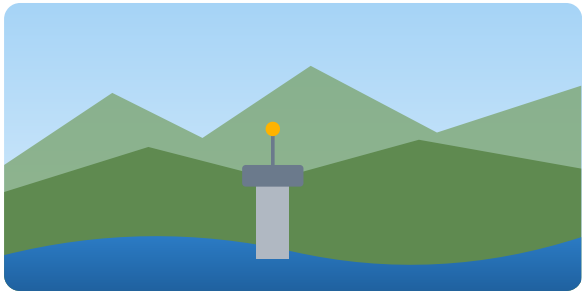


Monthly values as stored in 12.2 / 12.3

District Information

Alfred Nzo

● Live · as of Oct 2026



Status	Excellent
Province	Eastern Cape
WQI (Oct 2026)	91
Electrical conductivity (Oct 2026)	67 mS/m
pH (Oct 2026)	7.1
Turbidity (Oct 2026)	6.3 NTU
Dissolved oxygen (Oct 2026)	6.7 mg/L
Nitrate (NO_3^--N) (Oct 2026)	1.44 mg/L
Orthophosphate (PO_4-P) (Oct 2026)	0.050 mg/L
E. coli (Oct 2026)	95 cfu/100 mL

Latest Results (Selected District)

● Live · as of Oct 2026

Parameter	Value	Unit
pH	7.080	pH units
Electrical conductivity	66.67 mS/m	mS/m
Turbidity	6.33 NTU	NTU
Dissolved oxygen	6.660 mg/L	mg/L
BOD_5	1.750 mg/L	mg/L
COD	8.760 mg/L	mg/L
Nitrate (NO_3^--N)	1.440 mg/L	mg/L
Ammonia (NH_3-N)	0.125 mg/L	mg/L

Scroll sideways for more columns →

Hotspots / Non-compliance Locations

● Live · as of Oct 2026

#	District	Water body	Param
1	Joe Gqabi	—	Ortho (PO_4^{3-})
2	Sedibeng	—	Ortho (PO_4^{3-})
3	Vhembe	—	Ortho (PO_4^{3-})
4	Namakwa	—	Ortho (PO_4^{3-})
5	iLembe	—	Ortho (PO_4^{3-})
6	Sekhukhune	—	Ortho (PO_4^{3-})
7	Mangaung	—	Ortho (PO_4^{3-})
8	John Taolo Gaetsewe	—	Ortho (PO_4^{3-})

Scroll sideways for more columns →

Not available: Water body *(No water-body type (river, dam, wetland, groundwater) in the database: one record per district)*

Monitoring Districts by Water Body Type

Not yet available

No water-body type (river, dam, wetland, groundwater) in the database: one record per district

Database variables

Every table and column of the national database for this page, with the database's own names and units.

Spec coverage

● Live · as of Oct 2026

Water Quality: 4 tables · 35 columns · 35 live · 0 withheld · 0 duplicates Show every column →

Live columns are plotted below. Withheld columns failed a check or hold placeholder values (reason given); duplicates repeat another column.

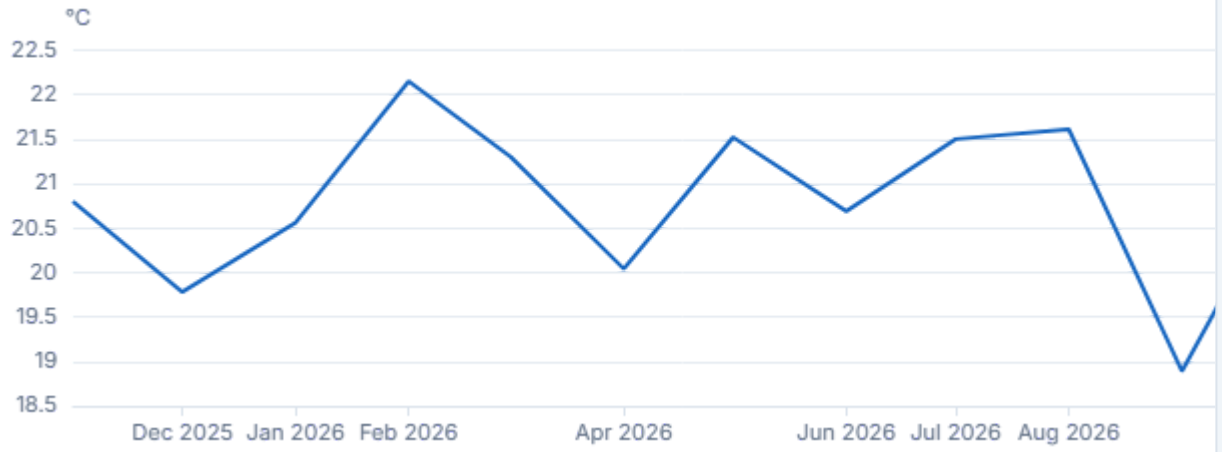
Physical Water Quality §12.1 `12.1_physical_water_quality_master`

7 live · 0 withheld · 0 duplicates data to Oct 2026

Physical Water Quality · Temperature (°C) · §12.1

● Live · as of Oct 2026

Temperature (°C)



Physical Water Quality · Turbidity (NTU) · §12.1

● Live · as of Oct 2026

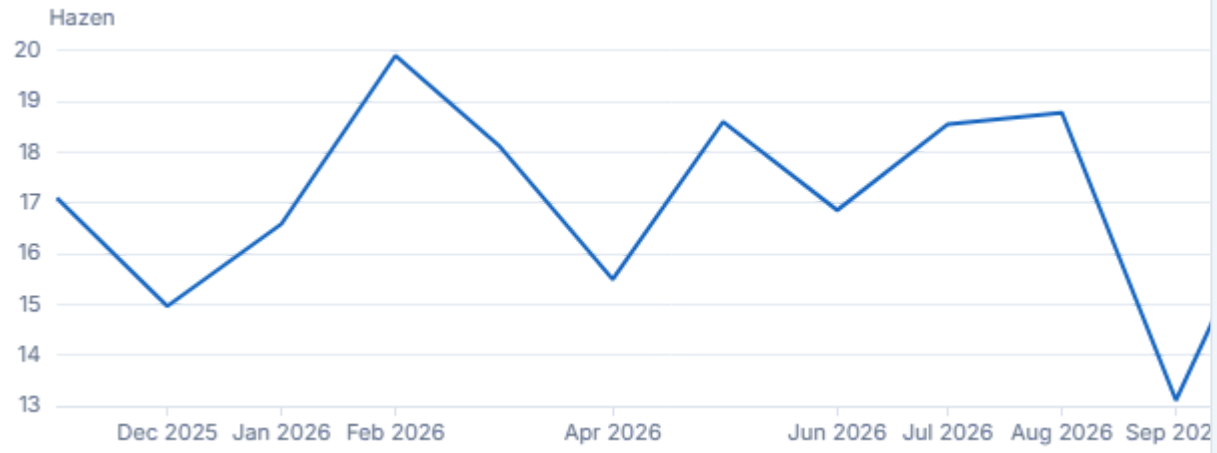
Turbidity (NTU)



Physical Water Quality · Colour (Hazen) · §12.1

● Live · as of Oct 2026

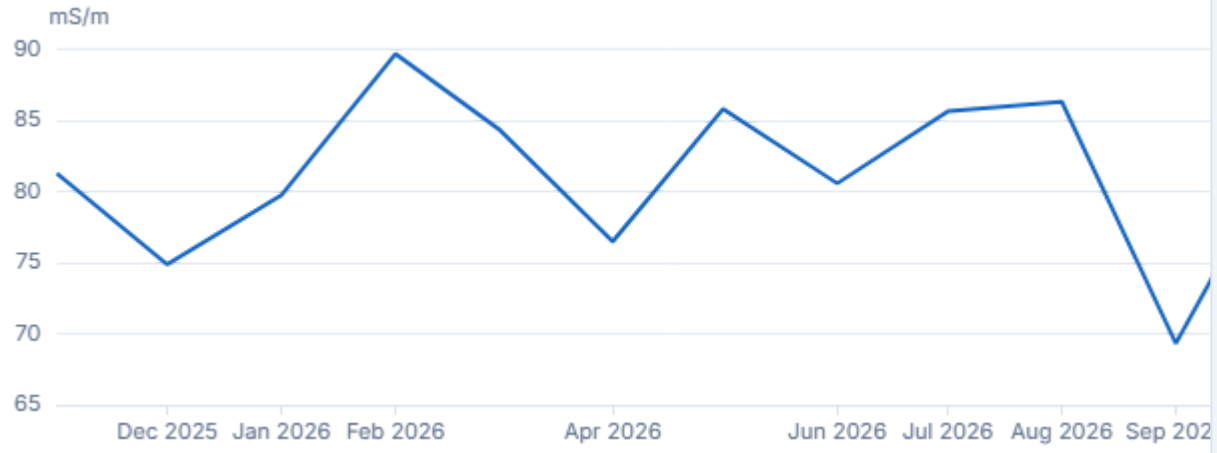
Colour (Hazen)



Physical Water Quality · Electrical Conductivity (EC) (mS/m) · §12.1

● Live · as of Oct 2026

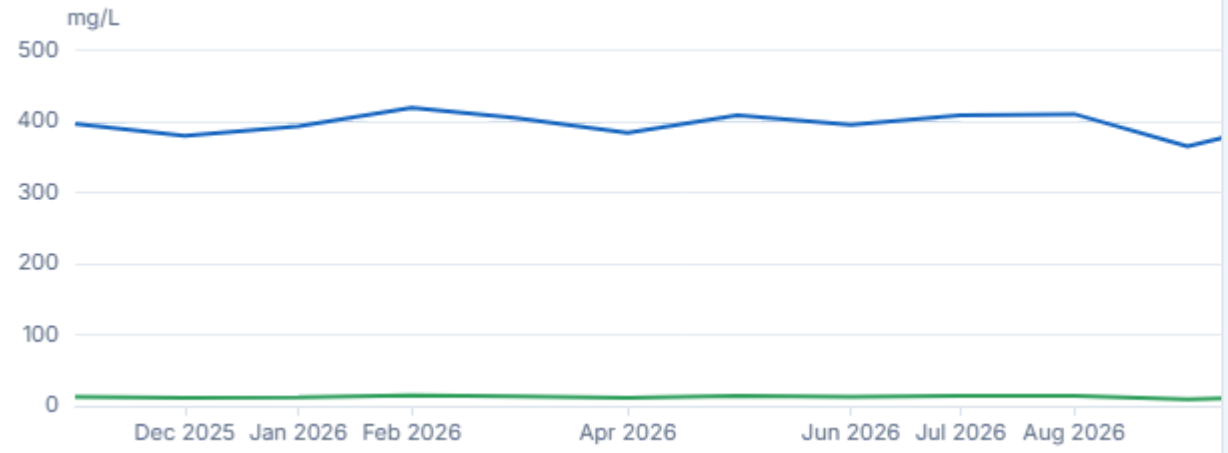
Electrical Conductivity (EC) (mS/m)



Physical Water Quality · Total (mg/L) · §12.1

● Live · as of Oct 2026

Total Dissolved Solids (TDS) (mg/L) | Total Suspended Solids (TSS) (mg/L) | All | Reset



Physical Water Quality · Salinity (mg/L) · §12.1

● Live · as of Oct 2026

Salinity (mg/L)

Chemical Water Quality §12.2 `12.2_chemical_water_quality_master`

16 live · 0 withheld · 0 duplicates data to Oct 2026

Chemical Water Quality · pH · §12.2

● Live · as of Oct 2026

pH

Chemical Water Quality · Dissolved Oxygen (DO) / BOD5 / COD / Ammonia (NH3-N) / Nitrate (NO3-N) / Nitrite (NO2-N) / Total Phosphorus (TP) / Orthophosphate (PO43-) / Iron/Manganese / TOC / Sulphate / Chloride / Fluoride (mg/L) · §12.2

● Live · as of Oct 2026

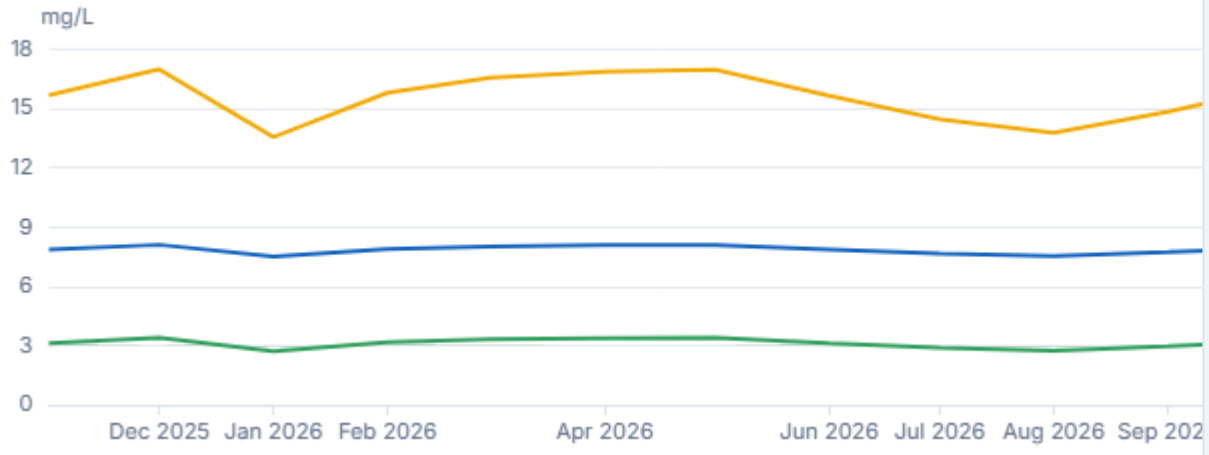
Dissolved Oxygen (DO) (mg/L) | BOD5 (mg/L) | COD (mg/L)

Ammonia (NH3-N) (mg/L) | Nitrate (NO3-N) (mg/L) | Nitrite (NO2-N) (mg/L)

Total Phosphorus (TP) (mg/L) | Orthophosphate (PO43-) (mg/L)

Iron/Manganese (mg/L) | TOC (mg/L) | Sulphate (mg/L) | Chloride (mg/L)

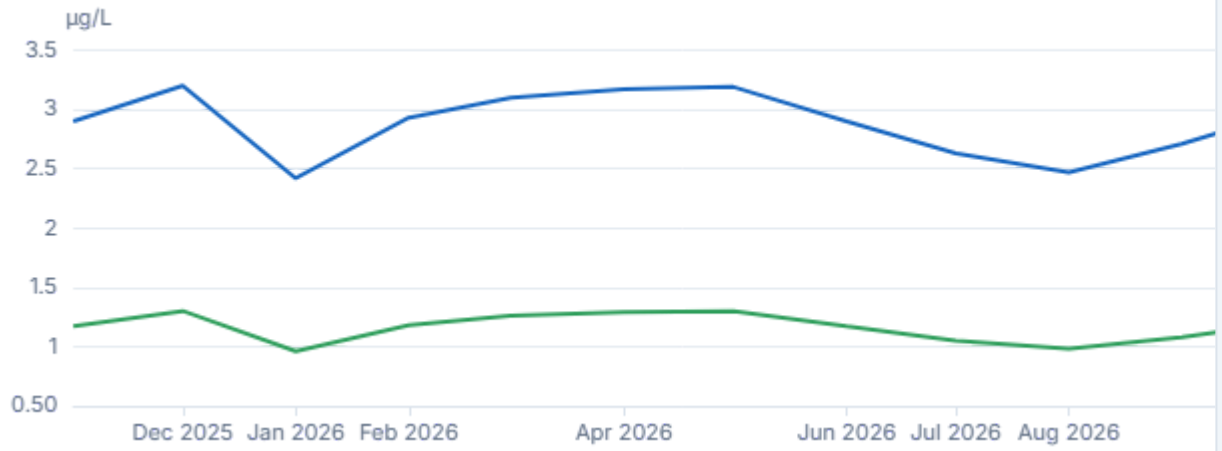
Fluoride (mg/L) All Reset



Chemical Water Quality · Lead/Cadmium/Chromium / Arsenic/Mercury (µg/L) · §12.2

Live · as of Oct 2026

Lead/Cadmium/Chromium (µg/L) | Arsenic/Mercury (µg/L) | All | Reset



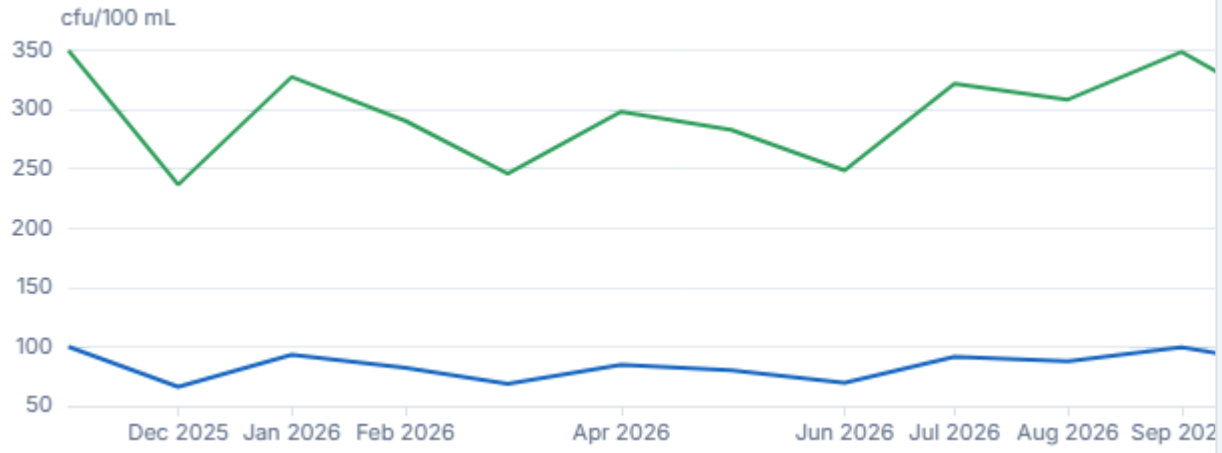
Biological Water Quality §12.3 `12.3_biological_water_quality_master`

7 live · 0 withheld · 0 duplicates data to Oct 2026

Biological Water Quality · E. Coli / Total Coliforms (cfu/100 mL) · §12.3

Live · as of Oct 2026

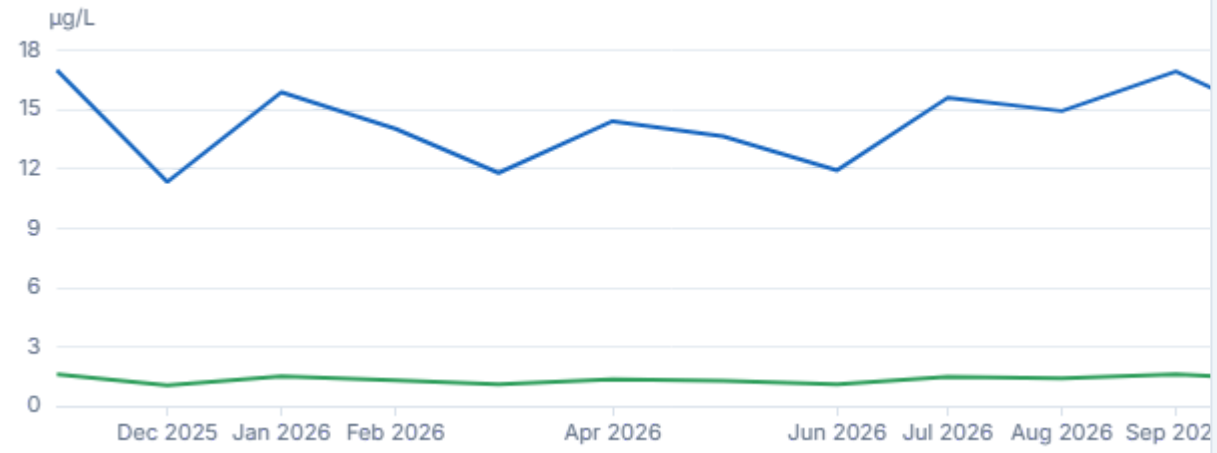
E. Coli (cfu/100 mL) | Total Coliforms (cfu/100 mL) | All | Reset



Biological Water Quality · Chlorophyll-a / Microcystin (µg/L) · §12.3

● Live · as of Oct 2026

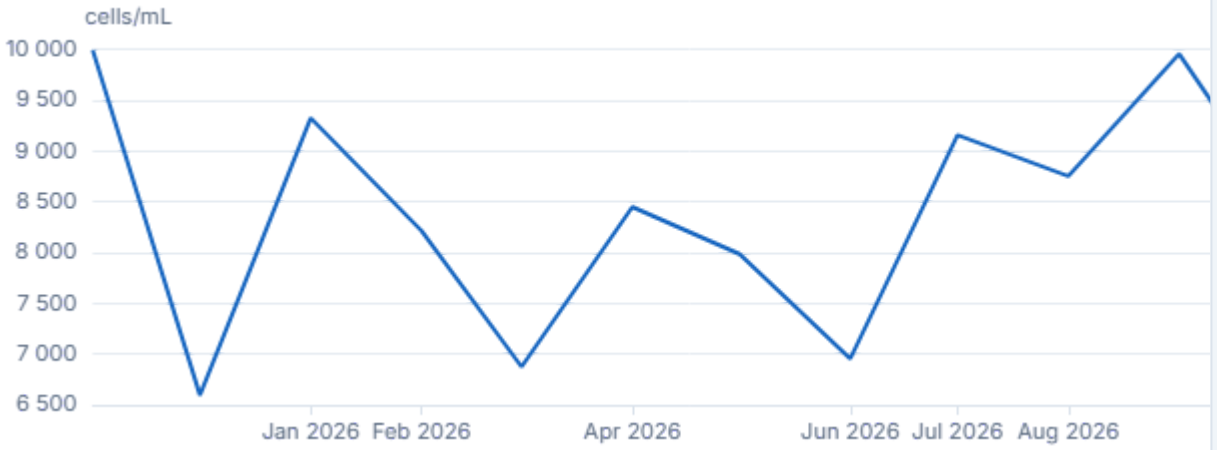
Chlorophyll-a (µg/L) | Microcystin (µg/L) | All | Reset



Biological Water Quality · Blue-green Algae (cells/mL) · §12.3

● Live · as of Oct 2026

Blue-green Algae (cells/mL)



Biological Water Quality · Macroinvertebrate Index (SASS5) · §12.3

● Live · as of Oct 2026

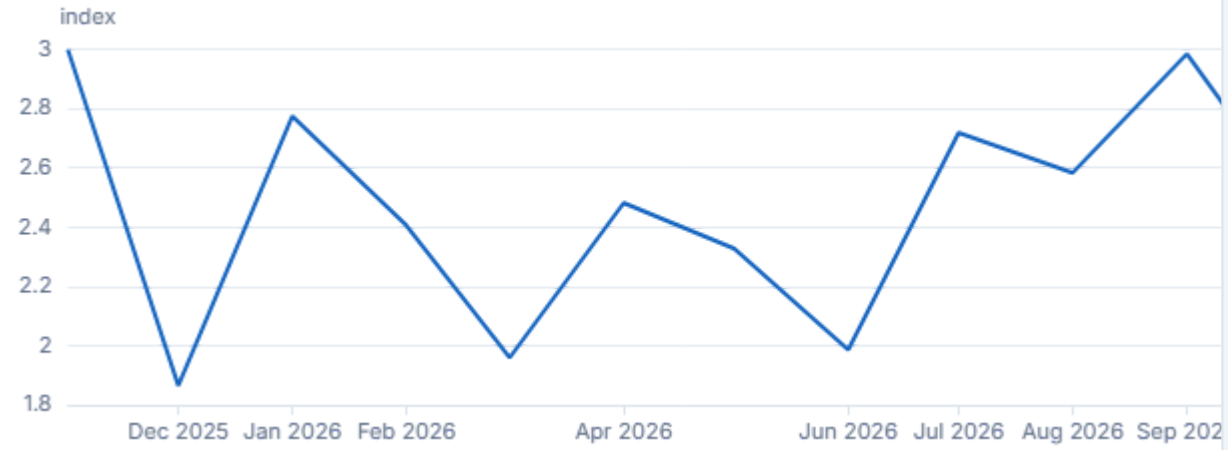
Macroinvertebrate Index (SASS5)



Biological Water Quality · Pathogens · §12.3

● Live · as of Oct 2026

Pathogens



SANSA Water Quality §12.4

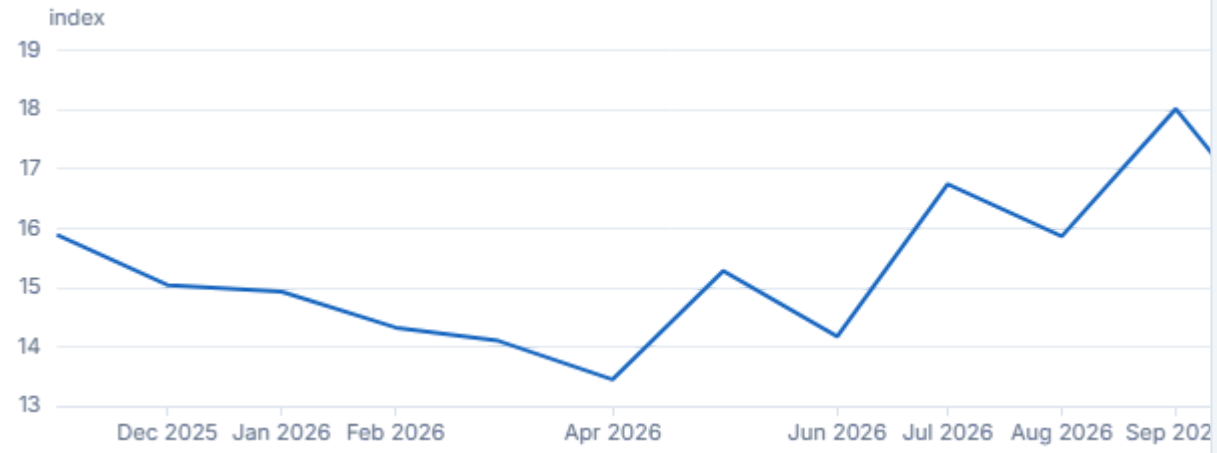
`12.4_sansa_water_quality_master` 5 live · 0 withheld · 0 duplicates

data to Oct 2026

SANSA Water Quality · Turbidity · §12.4

● Live · as of Oct 2026

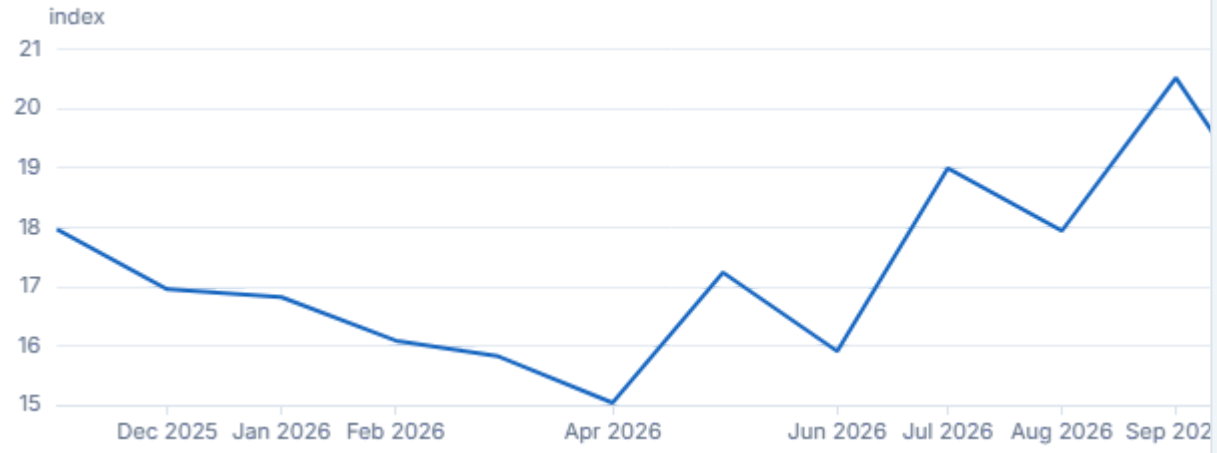
Turbidity



SANSA Water Quality · Chlorophyll-a · §12.4

● Live · as of Oct 2026

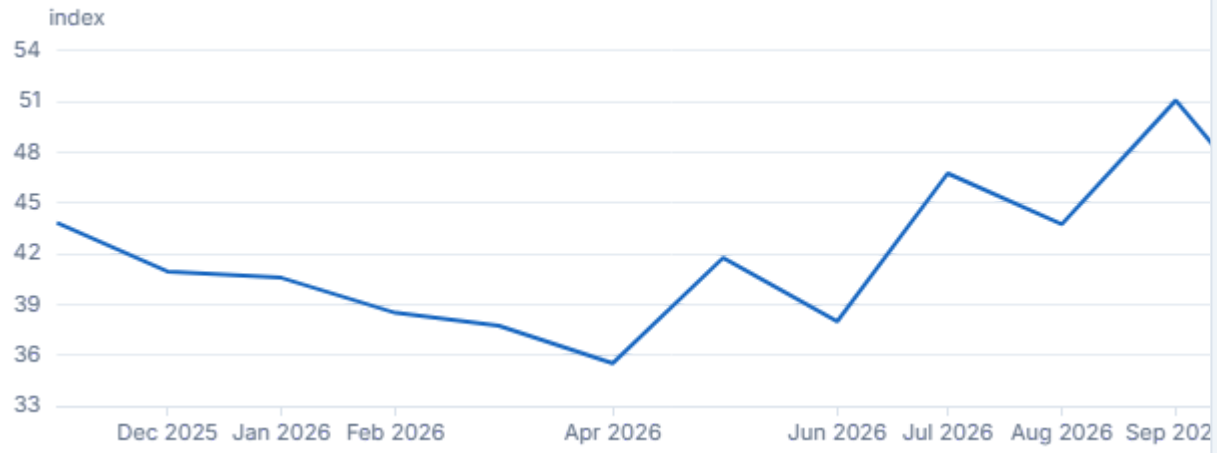
Chlorophyll-a



SANSA Water Quality · Cyanobacterial Bloom · §12.4

Live · as of Oct 2026

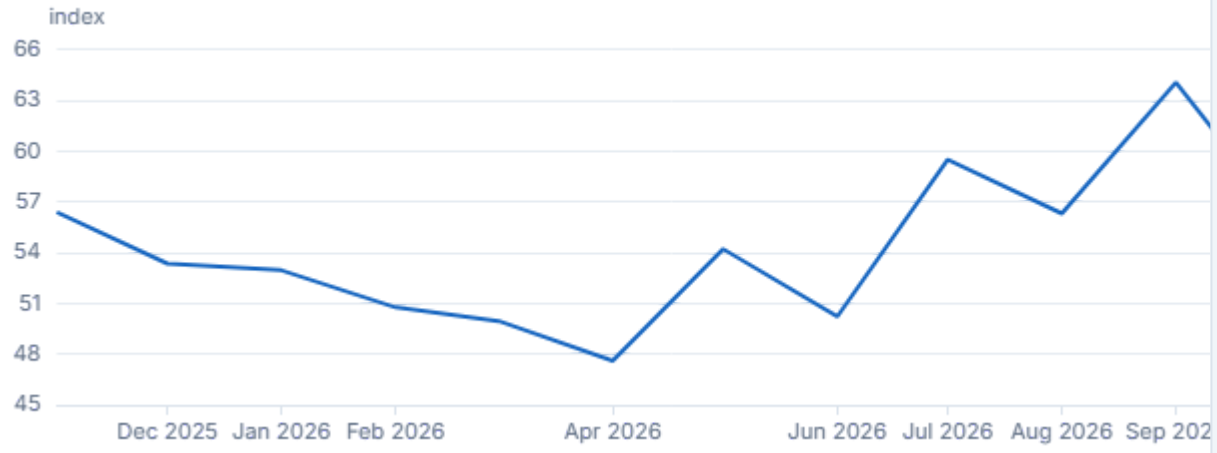
Cyanobacterial Bloom



SANSA Water Quality · CDOM · §12.4

Live · as of Oct 2026

CDOM



SANSA Water Quality · Total Suspended Matter (TSM) · §12.4

● Live · as of Oct 2026

Total Suspended Matter (TSM)

